We have, $x(m-n)+y(n-l)+z(l-m)$

$= a[(m-n)+(n-l)+(l-m)]+d[(m-n)(l-1)+(n-l)(m-1)+(l-m)(n-1)]$

$= a[0]+d[lm-ln-m+n+mn-lm-n+l+ln-mn-l+m]$

$= a(0)+d(0)=0$

(ii) On subtracting equation (2) from equation (1), equation (3) from equation (2) and equation (1) from equation (3), we get

$$x-y=(l-m)d$$

$$y-z=(m-n)d$$

$$z-x=(n-l)d$$

$$(x-y)n+(y-z)l+(z-x)m = [(l-m)n+(m-n)l+(n-l)m]d$$

$$= \left[ln-mn+lm-nl+nm-lm\right]d = 0$$

**Note**

In an Arithmetic Progression

- If every term is added or subtracted by a constant, then the resulting sequence is also an A.P.
- If every term is multiplied or divided by a non-zero number, then the resulting sequence is also an A.P.
- If the sum of three consecutive terms of an A.P. is given, then they can be taken as $a-d$, $a$ and $a+d$. Here the common difference is $d$.
- If the sum of four consecutive terms of an A.P. is given then, they can be taken as $a-3d$, $a-d$, $a+d$ and $a+3d$. Here common difference is $2d$.

**Example 2.29** In an A.P., sum of four consecutive terms is 28 and the sum of their squares is 276. Find the four numbers.

***Solution*** Let us take the four terms in the form $(a-3d), (a-d), (a+d)$ and $(a+3d)$.

Since, sum of the four terms is 28,

$$a-3d+a-d+a+d+a+3d=28$$

$$4a=28 \Rightarrow a=7$$

Similarly, since sum of their squares is 276,

$$(a-3d)^2+(a-d)^2+(a+d)^2+(a+3d)^2=276.$$

$$a^2-6ad+9d^2+a^2-2ad+d^2+a^2+2ad+d^2+a^2+6ad+9d^2=276$$

$$4a^2+20d^2=276 \Rightarrow 4(7)^2+20d^2=276.$$

$$d^2=4 \Rightarrow d=\pm\sqrt{4} \text{ then, } d=\pm 2$$

If $d=2$ then the four numbers are $7-3(2)$, $7-2$, $7+2$, $7+3(2)$

That is the four numbers are 1, 5, 9 and 13.



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